A smoothed Frobenius determinant bound for the tenth discriminant

Version 0.6.1: $q < 570$

Research note prepared with Codex

Revised exposition, 21 October 2026

Abstract

We derive $q < 570$ for the absolute fundamental discriminant of an imaginary quadratic field of class number one. An auxiliary inert prime makes a derivative interpolation determinant nonzero. Comparing its prime divisibility with its complex size forces a lower bound for a smoothed sum over split and ramified primes. Positive averaging and the quadratic norm Euler product give an incompatible upper bound. Five exact rational certificates make this comparison uniform on $570 \le q \le 2000$; the previously audited v5 result excludes $q \ge 2000$. The determinant arguments are proved here, using standard quadratic ideal theory, explicit ordinary-prime estimates, and classical analysis. No quasi-Riemann hypothesis or quadratic zero-free theorem is used.

1 Introduction

For a negative fundamental discriminant D, put $q = |D|$ and $K = \mathbb{Q}(\sqrt{D})$. We address Tao's request for tenth-discriminant bounds obtained without the stronger zero-location machinery discussed in the comments to his challenge [1, 2].

Theorem 1. *If $h(K) = 1$, then $q < 570$.*

The previously audited v5 result [12] gives $h(K) = 1 \Rightarrow q < 2000$. We use it only to dispose of $q \ge 2000$. The purpose of this paper is to exclude the remaining interval $[570, 2000]$. All estimates for that interval are developed below, relative to the stated ordinary-prime bounds and classical background. Section 8.2 then treats the finite range below 570.

Plan of the proof. Class number one gives two competing descriptions of split primes. Each split prime is a value of the quadratic norm form, which restricts how many there can be. On the other hand, a determinant argument forces a weighted sum of split and ramified primes to be large. The proof makes these descriptions incompatible in three steps.

1. *Construct a nonzero determinant.* Norm counts and ordinary-prime estimates supply an inert prime at each required scale. Reduction modulo that prime proves that a derivative interpolation matrix has a nonzero minor (Sections 2–3).

2. *Force a large prime sum.* Frobenius gives a lower bound for the minor's prime divisibility, while its entries give an upper bound for its complex size. Their comparison bounds a smoothed noninert-prime sum $\mathcal{T}(X)$ from below. The same sum also enters factorial divisibility, so a finite sequence of refinements strengthens the bound (Sections 4–6).

3. *Obtain a contradictory upper bound.* Average the lower bounds with a positive weight. The Euler product of quadratic norms bounds that very average from above. Explicit Poisson errors and rational interval calculations certify a positive gap on five bands covering $[570, 2000]$ (Sections 7–8).

The weight in $\mathcal{T}$ comes from an exact floor identity in the divisibility calculation. It is retained throughout the argument, so the determinant, factorial refinements, and final averaging all concern the same quantity. Section 6 assembles the individual estimates into the lower bound used in the last step.

Notation and background. All logarithms are natural, and $x_+ = \max(x, 0)$. A rational prime is called *noninert* when it is split or ramified in K. A *band* is one of the fixed intervals $[Q_0, Q_1]$ on which the estimates are uniform. The determinant is a theoretical object; the accompanying programs check its numerical inequalities and the final finite sieve. Quadratic ideal theory, ordinary-prime estimates of Rosser and Schoenfeld, and classical Poisson and special-function formulas are the external inputs.

2 Arithmetic and ordinary-prime estimates

2.1 The norm obstruction and lattice geometry

Assume henceforth that $h(K) = 1$ and $570 \le q \le 2000$. We first use the classical small-prime obstruction emphasized by Silberman in the discussion [3]. Every nonrational algebraic integer has norm at least $q/4$. A noninert rational prime has an integral ideal of norm p; under class number one its generator is nonrational. Consequently

$$p < q/4 \quad \Longrightarrow \quad p \text{ is inert.} \tag{1}$$

In particular q is an odd prime: an even discriminant has a small ramified prime 2, and a composite odd discriminant has a ramified prime at most $\sqrt{q} < q/4$. Inertness of 2 gives $q \equiv 3 \pmod 8$. The only units are ± 1, and

$$\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\omega, \qquad \omega = (1 + i\sqrt{q})/2, \qquad Q_q(a, b) = |a + b\omega|^2 = (a + b/2)^2 + qb^2/4.$$

Write

$$v = \sqrt{q}/2, \qquad \rho = \tfrac{1}{2}\sqrt{1 + v^2}.$$

The lattice has covolume v and a rectangular tiling of covering radius ρ. Its disk count $N(r)$ therefore satisfies

$$\frac{\pi \max(r - \rho, 0)^2}{v} \le N(r) \le \frac{\pi(r + \rho)^2}{v}. \tag{2}$$

Every principal ideal of norm m is a rotation and dilation by $\sqrt{m}$ of this lattice.

2.2 Explicit bounds for ordinary primes

The sieve below compares the logarithmic weight of all rational primes with the weight of split primes represented by the norm form. We record the ordinary-prime estimates needed for this

comparison and for the later valuation bounds. Put $\vartheta(x) = \sum_{p \le x} \log p$ and $A(x) = \sum_{p \le x} \log p/p$. We use Rosser–Schoenfeld [11], with $c_0 = 1.01624$:

$$\vartheta(x) < c_0 x, \tag{3}$$

$$x\left(1 - \frac{1}{2\log x}\right) < \vartheta(x) \quad (x \ge 563), \tag{4}$$

$$\vartheta(x) < x\left(1 + \frac{1}{2\log x}\right) \quad (x > 1), \tag{5}$$

$$A(x) > \log x - 1.334 - \frac{1}{2\log x} \quad (x \ge 319). \tag{6}$$

The last constant follows from their Theorem 6 and the explicit prime sum at 319: the certificate checks $\log 319 - A(319) + 1/(2\log 319) < 1.334$.

For sharper bounds at our scales, Table I on printed page 90 gives pairs (b, ϵ_b) such that $(1-\epsilon_b)x < \psi(x) < (1+\epsilon_b)x$ for $x \ge e^b$. The corollary to their Theorem 14 gives $0 \le \psi(x) - \vartheta(x) < 1.02\sqrt{x} + 3x^{1/3}$. Thus, uniformly for $\log x \ge m \ge b$,

$$1 - \epsilon_b - 1.02e^{-m/2} - 3e^{-2m/3} < \frac{\vartheta(x)}{x} < 1 + \epsilon_b. \tag{7}$$

Let $c(m), d(m)$ be any certified upper and lower bounds obtained by taking the best of these inequalities and (3)–(5). The table entries used are included as exact decimals in `parameters.json`.

We also use $A(R) < 2\log R$ at our scales. This follows without a further prime estimate: $\log(R!) \ge RA(R) - \vartheta(R)$, so $A(R) \le \log R + c_0 < 2\log R$.

3 Constructing the determinant

The first task is to make the determinant nonzero. We will use an inert prime as one of its dimensions, so we need such a prime in a controlled interval at every scale required by the certificate. The following sieve implements Pratt's suggestion to obtain the auxiliary prime from norm counts [7].

3.1 An auxiliary inert prime

The sieve counts possible split primes in an annulus. Removing elements divisible by a fixed collection of small ideals improves this count; a second-order inclusion-exclusion bound keeps the estimate explicit. Comparing it with the total prime weight will then force an inert prime.

Fix one band $Q_0 \le q \le Q_1$. Form the fixed, finite list of pairs

$$b > 0, \quad 2a + b > 0, \quad \gcd(a, b) = 1, \quad Q_{Q_1}(a, b) < (Q_0/4)^2, \tag{8}$$

ordered by $(Q_{Q_1}(a,b), a, b)$. Write $p_j(q) = Q_q(a_j, b_j)$, $p_j^- = Q_{Q_0}(a_j, b_j)$, and $p_j^+ = Q_{Q_1}(a_j, b_j)$.

Each $p_j(q)$ is prime. A composite norm below $(q/4)^2$ has a prime factor below $q/4$; inertness would force that prime to divide both coordinates. It is split rather than ramified: generators of the unique prime ideal above q have zero real part. These primes are distinct, because the four generators of a split prime norm are $\alpha, -\alpha, \bar{\alpha}, -\bar{\alpha}$, and the chosen quadrant contains exactly one of them.

Choose the first n such split primes, and the first h rational primes $r > 7$ with $r < Q_0/4$. Sieve by the two prime ideals above each selected p_j, and by each ideal $r\mathcal{O}_K$. Their norms are p_j, p_j, r^2, and they are pairwise coprime. Also impose the exact condition $\gcd(a, b, 210) = 1$.

If an element belongs to k of the selected ideals, then $1-k+\binom{k}{2}$ is an upper bound for the indicator that it belongs to none. Apply this second Bonferroni inequality, multiplied by the exact 210 indicator, to the annulus $U<|\alpha|^2\le\kappa U$. A split prime in this interval has four generators. Assume $U>Q_1$ and $U>\max_{j\le n}p_j^+$, so the sieving primes themselves are not counted.

For an ideal of norm m, the disk estimate gives an annulus count with area term $\pi(\kappa-1)U/(vm)$, boundary error at most $2\pi\rho(1+\sqrt{\kappa})\sqrt{U}/(v\sqrt{m})$, and constant error at most $2\pi\rho^2/v$. The 210 expansion has

$$\sum_{d\mid 210}\frac{\mu(d)}{d^2}=\frac{768}{1225},\qquad \sum_{d\mid 210}\frac{|\mu(d)|}{d}=\frac{96}{35},\qquad \sum_{d\mid 210}|\mu(d)|=16.$$

Define

$$u=2\sum_{j\le n}(p_j^+)^{-1}+\sum_r r^{-2},\qquad u_2=2\sum_{j\le n}(p_j^+)^{-2}+\sum_r r^{-4},$$
$$z=2\sum_{j\le n}(p_j^-)^{-1/2}+\sum_r r^{-1},\qquad z_2=2\sum_{j\le n}(p_j^-)^{-1}+\sum_r r^{-2},\qquad m_0=2n+h.$$

The certificate checks that the analogous u at the minus endpoints is less than 1. The area polynomial then decreases in every reciprocal ideal norm, since its partial derivative is -1 plus the sum of the other reciprocals. Therefore uniform upper bounds for the number of split primes in $(U,\kappa U]$ are

$$A_*(\kappa-1)U+B_*(1+\sqrt{\kappa})\sqrt{U}+C_*, \tag{9}$$

where $v_-=\sqrt{Q_0}/2$, $v_+=\sqrt{Q_1}/2$, and

$$A_*=\frac{\pi}{4v_-}\frac{768}{1225}\left(1-u+\frac{u^2-u_2}{2}\right),$$
$$B_*=\frac{\pi}{4}\sqrt{1+v_-^{-2}}\frac{96}{35}\left(1+z+\frac{z^2-z_2}{2}\right),$$
$$C_*=2\pi(v_++v_+^{-1})\left(1+\frac{m_0(m_0+1)}{2}\right).$$

For a parameter row, put $M=2^a$, and let $2^a\le U\le 2^b$. Set $d=d(\log M)$, $c=c(\log M)$. An inert prime exists in $(U,\kappa U]$ if

$$G(x)=\kappa d-c-(x+\log\kappa)\{A_*(\kappa-1)+B_*(1+\sqrt{\kappa})e^{-x/2}+C_*e^{-x}\}>0,\quad x=\log U. \tag{10}$$

Indeed, the total prime logarithmic weight exceeds the possible split weight, and the ramified prime is below the interval. Since $x+\log\kappa>4$, the function G is concave. It suffices to check its two endpoints. These are exact rational interval checks in the certificate; all κ,n,h,a,b are in the parameter file.

3.2 The derivative matrix and its rank

We use Green's derivative matrix [4], with interpolation points in a disk as suggested by Pratt [7]. Choosing the auxiliary prime itself as the monomial width makes the finite-field degree count particularly simple. Take an inert prime $p_0\in(U,\kappa U]$, and set $R=p_0$. Throughout the proof use the fixed constants

$$t=\frac{3}{64},\qquad C=\frac{27}{64},\qquad S=\lfloor tR\rfloor,\qquad J=\lceil R/9\rceil,\qquad T=\sqrt{CRv/\pi}+\rho. \tag{11}$$

Use all n lattice points z_i in the disk of radius T. From this point on, n denotes this lattice-point count, rather than the number of selected split primes in the sieve. Then $n \ge CR$, so $nJ \ge tR^2 \ge RS$. The matrix has rows $0 \le r < R, 0 \le s < S$, columns (i,k), $0 \le k < J$, and entries

$$\partial_z^k (z^r w^s)\big|_{(z_i, \bar{z}_i)}.$$

The next lemma is van Dijk's auxiliary-prime argument [5]. Green also attributes a related unpublished argument to Pescador [6]; the proof here uses the public finite-field argument.

Lemma 2. *If $2T < R$ and $J \le R$, this matrix has full row rank.*

Proof. Reduce modulo $p_0\mathcal{O}_K$, a field of p_0^2 elements. The points are distinct: a nonzero element of $p_0\mathcal{O}_K$ has modulus at least p_0, whereas two points in the disk differ by less than $2T$. Frobenius gives $\bar{z}_i = z_i^{p_0}$. If a nonzero row combination vanished, substitute $w = z^{p_0}$ in its polynomial. The exponents $r + p_0 s$ are distinct, so the result is a nonzero polynomial of degree at most $RS - 1$. For $k < J \le p_0$, its ordinary derivatives agree with the specified z-derivatives in characteristic p_0. Each node is thus a zero of multiplicity at least J, since $k! \ne 0$ for $k < p_0$. This gives at least $nJ \ge RS$ zeros, a contradiction. Full rank after reduction implies full rank in characteristic zero. □

Fix a nonzero RS-square minor $\mathcal{D}$, and put $N = RS$ and $\Delta = |\mathcal{D}|^2 \in \mathbb{Z}_{>0}$. Different triangular row changes below may be used for different primes: they leave this same determinant unchanged.

4 Prime divisibility of the determinant

We now bound the prime divisibility of the fixed positive integer Δ. The baseline bound comes from Frobenius at inert primes. Its leading term is a sum over all primes minus a smoothed sum over noninert primes. Higher Frobenius factors and factorials add compatible divisibility to the same minor. The resulting estimates will be combined with the size bound in Section 6.

4.1 The baseline bound and its smoothing weight

The floor $\lfloor r/p \rfloor$ counts how many Frobenius factors can be taken from a row of degree r. Summing this floor over the rows produces the following weight. For $0 < u \le 1$, put

$$f(u) = 2au - a(a+1)u^2, \qquad a = \lfloor 1/u \rfloor; \qquad f(u) = 0 \ (u > 1), \quad f(0) = 1. \tag{12}$$

This is continuous, nonnegative, nonincreasing, and at most 1. Also

$$f(u) = 2u \sum_{m \ge 1} (1 - mu)_+, \qquad \frac{2}{R^2} \sum_{r=0}^{R-1} \left\lfloor \frac{r}{p} \right\rfloor = \frac{f(p/R)}{p}. \tag{13}$$

Define the noninert sum

$$\mathcal{T}(X) = \sum_{p \text{ noninert}} \frac{\log p}{p} f(p/X). \tag{14}$$

It is nonnegative and nondecreasing in X, and is at most the ordinary cumulative noninert sum.

Green's triangular change [4], using $a_{r,s} = \min(\lfloor r/p \rfloor, S - 1 - s)$, replaces $z^r w^s$ by $z^{r - p a_{r,s}} (z^p - w)^{a_{r,s}} w^s$. For inert p, every ordinary derivative of $z^p - w$ is divisible by p, so

$$v_p(\Delta) \ge 2 \sum_{r,s} a_{r,s}.$$

Write this normalized valuation as $e(R,S,p)$. Counting layers gives

$$e(R,S,p)=\frac{2}{R^2S}\sum_{h=1}^{S-1}(S-h)(R-ph)_+. \tag{15}$$

For $p(S-1)\le R-1$, this equals $(S-1)/R-p(S^2-1)/(3R^2)$. In general,

$$e(R,S,p)\ge \frac{f(p/R)}{p}-\frac{R-1}{2RSp}-\frac{(R-1)(2R-1)}{6RSp^2}. \tag{16}$$

Indeed, sum $\sum_{s<S}\min(k,S-1-s)\ge Sk-k(k+1)/2$, then use $k(k+1)\le r^2/p^2+r/p$ and (13).

The main term in the inert-prime sum in (16) is exactly

$$\sum_p \frac{\log p}{p}f(p/R)-\mathcal{T}(R).$$

Thus the prime sum that must be bounded from below is $\mathcal{T}(R)$. Its specific weight will also allow the factorial gains and the final norm-series comparison to be expressed in terms of $\mathcal{T}$.

4.1.1 The all-prime term

We first compare the sum over all primes with $A(R)$, retaining a nonnegative correction from the floor remainder. If $R=ap+b$, the exact floor identity gives

$$\frac{f(p/R)}{p}=\frac{1}{p}-\frac{1}{R}+\frac{1}{R}\delta_a(p/R),\qquad \delta_a(u)=2a+1-u^{-1}-a(a+1)u.$$

On $[1/(a+1),1/a]$, δ_a is nonnegative, vanishes at the endpoints, and has one maximum. Partial summation with $d_*x\le\vartheta(x)\le c_*x$, valid for $x\ge M/33$, gives

$$G_{\text{floor}}=\max(0,Bd_*-Ac_*),\quad A=\frac{\log 33-H_{33}+1}{2},\quad B=\frac{H_{32}-\log 33}{2}. \tag{17}$$

Here $H_n=\sum_{j\le n}1/j$. The increasing and decreasing pieces contribute integrals $\frac{1}{2}\log(1+1/a)-\frac{1}{2(a+1)}$ and $\frac{1}{2a}-\frac{1}{2}\log(1+1/a)$, respectively; sum for $1\le a\le 32$. Thus

$$\sum_p\frac{\log p}{p}f(p/R)\ge A(R)-c(\log M)+G_{\text{floor}}\qquad (R\ge M). \tag{18}$$

4.1.2 The small primes and the quadratic tail

The loss involving p^{-2} in (16) is needlessly large for small primes. We evaluate their contribution directly, then remove the known split and ramified primes from the remaining inert-prime tail. Put $H=\lfloor 1/t\rfloor=21$,

$$a_H=\sum_{p\le H}\frac{\log p}{p},\quad \theta_H=\sum_{p\le H}\log p,\quad W_H=\sum_{p\le H}\log p\,(t-pt^2/3).$$

Let B_H be a certified upper bound for $\sum_{p>H}\log p/p^2$. Specifically, with $P=10000$, partial summation and (3) give

$$B_H=\sum_{H<p\le P}\frac{\log p}{p^2}+\frac{2c_0}{P}-\frac{\vartheta(P)}{P^2}>\sum_{p>H}\frac{\log p}{p^2}.$$

With all pairs in (8), define

$$K_2 = \sum_j \frac{\log p_j^+}{(p_j^+)^2} + \frac{\log Q_1}{Q_1^2}. \tag{19}$$

The summands decrease with the prime, so the inert quadratic tail is at most $B_H - K_2$. Combining (16), (18), the exact small-prime formula, and $A(R) < 2\log R$, gives

$$\begin{aligned}\frac{\log \Delta}{NR} \geq A(R) - c(\log M) + G_{\mathrm{floor}} - \mathcal{T}(R) - a_H + W_H \\ - \frac{B_H - K_2}{3(t-1/M)} - \frac{2\theta_H}{M} - \frac{\log M}{(t-1/M)M}.\end{aligned} \tag{20}$$

The next two subsections add further divisibility while preserving this bound for the same determinant.

4.2 Higher Frobenius factors

Kerr's suggestion to retain higher Frobenius relations, including the ninth-power relation at 3, motivates the extra factors here [9]. To add their gains, we must put them into one triangular row transformation and account for differentiation of the resulting product. At an inert prime use

$$G = z^p - w, \qquad H_1 = z^{p^2} - z, \qquad H_2 = w^{p^2} - w.$$

For each row let

$$a = \min(\lfloor r/p \rfloor, S - 1 - s), \quad c = \lfloor (r - pa)/p^2 \rfloor, \quad b = \lfloor s/p^2 \rfloor,$$

and replace the row polynomial by

$$z^{r-pa-p^2c} G^a H_1^c \, w^{s-p^2b} H_2^b. \tag{21}$$

Its leading term is $z^r w^s$; all other terms have smaller lexicographic index and remain in the rectangle, since their z-degree is at most r and their w-degree at most $s + a \leq S - 1$.

All three factors have values divisible by p. A Hasse derivative can lose a factor G only at order p, and a factor H_1 only at order 1 or p^2. At total order k, the number of lost factors is therefore at most

$$L = \min(k, c) + \left\lfloor \frac{k - \min(k, c)}{p} \right\rfloor.$$

Ordinary differentiation supplies $k!$. The elementary inequality

$$c + \lfloor k/p \rfloor - L \geq \lfloor c/p \rfloor$$

holds both for $k \geq c$ and $k < c$. Consequently the entry valuation is at least

$$a + b + \lfloor c/p \rfloor + \sum_{e \geq 2} \lfloor k/p^e \rfloor. \tag{22}$$

The additional residual row factor is $\lfloor (r - pa)/p^3 \rfloor$. Moreover, beyond this bound one may retain

$$\max\left(0, (1 - 1/p)(c - k) - 1\right). \tag{23}$$

For $k < c$, this follows by bounding $\lfloor k/p \rfloor$ below by $k/p - 1$; for $k \geq c$, (22) suffices.

4.2.1 Averaging the row and column gains

The minor contains every row but only a selection of the available columns. Each derivative order occurs in at most n columns. This restriction bounds the mean derivative order in any chosen minor; the parameter b_0 will be supplied by the lattice estimate in Section 5.2. Suppose $R \ge M$, put $\varepsilon = 1/M$, and choose $b_0 \le S/n$. At most n columns have each derivative order. Their mean order is at least $(b_0R - 1)/2$, and at most j_*R, where

$$j = 1/9 + \varepsilon, \qquad j_* = j - b_0/2 + b_0/M. \tag{24}$$

The latter is a conservative consequence of sorting the orders: the ith is at most $J-1-\lfloor (N-1-i)/n \rfloor$.

The b and higher-column terms in (22) give

$$G_{\mathrm{small}} = \sum_{p<Q_0/4} \log p \left\{ \sum_{\substack{e\ge 2 \\ p^e<M}} \left(\frac{b_0-\varepsilon}{p^e} - 2\varepsilon \right)_+ + \left(\frac{t-\varepsilon}{p^2} - 3\varepsilon \right)_+ \right\}. \tag{25}$$

For the residual factor, define

$$A_p(t) = \begin{cases} 1/(2p^2) - t/(2p) + t^2/6, & pt \le 1, \\ 1/(6tp^3), & pt \ge 1. \end{cases}$$

Then the further gains are

$$G_{\mathrm{res}} = \sum_{p<Q_0/4} \log p \left(\frac{2A_p(t)}{p} - (2 + 2/p^3)\varepsilon \right)_+, \tag{26}$$

$$G_{\mathrm{coup}} = 2 \sum_{p<Q_0/4} \log p \left((1-1/p)\{A_p(t) - (1+1/p^2)\varepsilon - j_*\} - \varepsilon \right)_+. \tag{27}$$

Here is a direct integral justification. Put $x_r = r/R$ and $y_s = (S-1-s)/R$. We have $c/R \ge (x_r - py_s)_+/p^2 - 1/R$. On the cell $[x_r, x_r + 1/R] \times [y_s, y_s + 1/R]$, $(x-py)_+ \le (x_r - py_s)_+ + 1/R$. The average of $(x-py)_+/p^2$ over $[0,1] \times [0, S/R]$ equals $A_p(S/R)$, which decreases as S/R increases. Thus the normalized mean of c is at least $A_p(t) - (1+1/p^2)/M$. For (26), use the exact identity $\lfloor c/p \rfloor = \lfloor (r - pa)/p^3 \rfloor$. The same cell argument, now divided by p^3, bounds its normalized mean below by $A_p(t)/p - (1+1/p^3)/M$. The mean of a positive part is at least the positive part of its mean, so (23) and (24) give (27). All these terms are added to (20).

4.3 Factorial divisibility and finite refinement

Noninert primes contribute factorial divisibility of their own. The floor identity (13) expresses its first-power contribution using the same function $\mathcal{T}$, now evaluated at smaller scales. Consequently an already proved lower bound for $\mathcal{T}$ strengthens the determinant comparison at the next stage. At a split prime, replace w^s by its falling factorial $(w)_s$. Since $\mathcal{O}_K/p^e\mathcal{O}_K \simeq (\mathbb{Z}/p^e\mathbb{Z})^2$, $p^{v_p(s!)}$ divides its value. The column factor $k!$ remains. At the ramified prime keep the column factorial only. These row changes are triangular, and therefore bound the same determinant.

Suppose a nonnegative nondecreasing function Φ is already known to satisfy $\mathcal{T}(e^x) \ge \Phi(x)$ throughout the band. The first-power part of the split row factorial contributes exactly $(S/R)\mathcal{T}_{\mathrm{split}}(S)$, by (13). It is therefore at least

$$(t-\varepsilon) \left(\Phi(\log(M(t-\varepsilon))) - \frac{\log Q_0}{Q_0} \right)_+.$$

For the columns put $K_0 = \lfloor b_0 R \rfloor$. The increasing chosen orders are at least $\lfloor i/n \rfloor$. Since $N/n \ge K_0$, the average of $\lfloor k/p \rfloor$ over these columns is at least its average for $0 \le k < K_0$. Using (13) again gives a contribution at least $(K_0/R)\mathcal{T}(K_0)$. Hence a compatible total gain is

$$G_{\mathrm{fb}} = (b_0 - \varepsilon)\Phi(\log(M(b_0 - \varepsilon))) + (t - \varepsilon)\left(\Phi(\log(M(t - \varepsilon))) - \frac{\log Q_0}{Q_0}\right)_+ . \tag{28}$$

We start with $\Phi = 0$, compute a new lower bound using the size estimate below, and repeat eight times. Section 6 describes the resulting finite induction. Each stage uses only a bound already proved at the preceding stage.

5 An upper bound for the complex size

We now bound the same nonzero minor from above. A typical entry contains both a falling factorial from differentiation and a power of the radius of its interpolation point. These depend on the same monomial exponent. The following estimate keeps that dependence when averaging over rows and columns. The result is (33); it will be compared with the divisibility bounds in Section 6.

5.1 A pointwise inequality for each entry

Fix $\tau = 3/10$, and, for $0 \le k \le u \le 1$, put

$$F(u,k) = u\log u - (u-k)\log(u-k) - k, \qquad \phi(u) = \tau(u\log u - u).$$

At zero use the continuous values. For $0 < a \le 1$, define $\psi(k,a) = \max_{k \le u \le 1}\{(u-k)\log a + F(u,k) - \phi(u)\}$. The maximized function is strictly concave. If $a \le 1-k$, its critical point satisfies $u - k = au^{1-\tau}$, and

$$\psi(k,a) = \tau u + (1-\tau)k\log u - k.$$

If $a \ge 1-k$, the maximum is at 1 and

$$\psi(k,a) = (1-k)\log\frac{a}{1-k} - k + \tau.$$

Differentiation shows that ψ decreases in k, increases in a, and is 1-Lipschitz in a. In particular, for $k \le j$,

$$\psi(k,a) + B \ge 0, \qquad B = (1-\tau)j(1-\log j).$$

To prepare for averaging over the disk and derivative orders, let $I(k) = \int_0^1 2a\psi(k,a)\,da$ and $K(j) = \int_0^j I(k)\,dk$. We have $I(0) = 2\tau^2/(2\tau+1)$. Changing variables from a to the critical point gives

$$I'(k) = -(1-\tau)\int_k^1 (u-k)^2 u^{2\tau-3}\,du + \log(1-k) + k - k^2/2.$$

Thus, writing $A = 2\tau$,

$$K(j) = j\frac{2\tau^2}{2\tau+1} - (1-\tau)\left\{\frac{j^2}{2A} - \frac{j^3}{3(A-1)} + \frac{j^4}{12(A-2)} - \frac{2j^{A+2}}{A(A-1)(A-2)(A+1)(A+2)}\right\} + \frac{j}{2} - \frac{3j^2}{4} + \frac{(1-j)^2}{2}\log(1-j) + \frac{j^3}{6} - \frac{j^4}{24}. \tag{29}$$

For an actual row and derivative order, take $u = (r+1)/R$ and $k' = k/R$. Increasing-log integration gives $\log(r)_k \le k \log R + RF(u, k')$. Clip the normalized radius to $a = \max(1, |z|)/T$. A nonzero entry satisfies

$$\log|\partial_z^k(z^r w^s)| \le (r+s)\log T + k\log(R/T) + R\{\phi(u) + \psi(k', a)\} + s\log a - \log a.$$

The final term is at most $\log T$. Since ϕ decreases, its right Riemann sum gives a normalized row contribution at most $-3\tau/2$ to $\log \Delta/(NR)$.

5.2 Averaging over the lattice and chosen columns

The minor may select any N of the available columns. We therefore need an upper bound that is uniform over that selection. Adding B makes the quantity to be summed nonnegative, so we can first sum over all available columns and then subtract NB.

For uniform estimates set $\varepsilon = 1/M$,

$$r_* = \sqrt{CMv_+/\pi}, \quad e \ge \rho_+/r_*, \quad \nu = C(1+2e)^2, \quad b_0 \le (t-\varepsilon)/\nu, \quad g = \tfrac{1}{2}\log(C/(2\pi)) + \log(1+e). \tag{30}$$

Here $v_+ = \sqrt{Q_1}/2$, $\rho_+ = \sqrt{1+v_+^2}/2$. Then $n \le \nu R$, $b_0 \le S/n$, and $\log T \le \frac{1}{2}\log R + \frac{1}{4}\log q + g$. The monotonicity of $\rho/\sqrt{v} = \frac{1}{2}\sqrt{v + v^{-1}}$ justifies use of the upper band endpoint.

Lattice-cell integration, using nonnegativity and the Lipschitz property, gives for each order

$$\sum_{|z|\le T} \{\psi(k/R, a_z) + B\} \le \nu R \left\{ I(k/R) + B + \frac{5e}{3(1+e)} \right\}.$$

Indeed, cells lie in the disk of radius $T + \rho$. Its normalized radial argument is $((1+2e)a + e)/(1+e)$, whose difference from a is $e(1+a)/(1+e)$. Integrating against $2a\,da$ gives the displayed error. Clipping at 1 only improves the estimate; $\rho \ge 1$ also covers the clipped origin.

We may fill in unchosen columns before subtracting NB. Since I decreases and $I + B \ge 0$, the order sum is at most

$$R\left\{ K(j) + j\left(B + \frac{5e}{3(1+e)} \right) \right\} + I(0) + B.$$

It follows that the normalized joint contribution is at most

$$D_* = -\frac{3\tau}{2} + \frac{2\nu}{t-\varepsilon}\left\{ K(j) + j\left(B + \frac{5e}{3(1+e)} \right) + \varepsilon(I(0) + B) \right\} - 2B. \tag{31}$$

5.3 The remaining radius factor and the final bound

The entry estimate also contains $s \log a$, which has not yet been used. It is nonpositive. To retain a definite saving, count how many chosen columns must lie inside each smaller disk and how small their assigned row exponents can be in a determinant term. At least $\lambda(a)N$ chosen columns have radius at most aT, where

$$\lambda(a) = \left(1 - \frac{jC}{t-\varepsilon}\{(1+2e)^2 - \max(a(1+e) - e, 0)^2\} \right)_+.$$

Among that many distinct rows the normalized sum of the s's is at least $((t-\varepsilon)\lambda(a)^2 - \varepsilon)/2$. Put

$$\delta = \frac{jC}{t-\varepsilon}\left\{ (1+2e)^2 + \frac{e}{2+e} \right\} - 1.$$

Since $\max(a(1+e)-e,0)^2 \geq a^2 - e/(2+e)$ and $jC/(t-\varepsilon) \geq 1$, we have $\lambda(a) \geq (a^2-\delta)_+$. The certificate checks $0<\delta<1$ and $\delta \geq e$, so $\sqrt{\delta} \geq 1/T$. Integrating from $\sqrt{\delta}$ to 1 gives the additional saving

$$G_s = \left((t-\varepsilon) \left\{ \frac{1}{4} - \delta + \frac{3\delta^2}{4} - \frac{\delta^2}{2} \log \delta \right\} + \frac{\varepsilon}{2} \log \delta \right)_+ . \tag{32}$$

There are at most $N!$ terms in the determinant. Using $\log N! \leq N \log N$, (24), and $1 < T < R$, we obtain

$$\frac{\log \Delta}{NR} \leq (1-\alpha)\log R + \frac{\beta}{4}\log q + \beta g + D_* + \frac{2\log M}{M} - G_s + \frac{2\log(tM^2)}{M}, \tag{33}$$

where $\alpha = (1-t)/2 - j_*$, $\beta = 1 + t - 2j_*$. The functions $(\log R)/R$ and $\log(tR^2)/R$ decrease on the certified ranges. The terms D_* and $-G_s$ account for different parts of the displayed entry bound: the former treats $\phi + \psi$, and the latter treats $s \log a$.

6 Assembling the lower bound for the prime sum

All the determinant estimates are now in place. The baseline divisibility bound has the form "ordinary-prime contribution minus $\mathcal{T}(R)$", with the compatible gains added to it. Since this cannot exceed the complex-size upper bound, rearranging forces $\mathcal{T}(R)$ to be large.

6.1 From a determinant scale to an interval of scales

Combine (20), (6), (25)–(27), (28), and (33). This gives

$$\mathcal{T}(R) \geq \alpha \log R - \frac{\beta}{4}\log q - \mathcal{K}, \tag{34}$$

where the fully explicit constant is

$$\begin{aligned} \mathcal{K} = {} & a_H - W_H + c(\log M) + \frac{B_H - K_2}{3(t-\varepsilon)} + \beta g + D_* + \frac{2\log M}{M} - G_s \\ & + \frac{2\theta_H}{M} + \frac{\log M}{(t-\varepsilon)M} + \frac{2\log(tM^2)}{M} + 1.334 + \frac{1}{2\log M} \\ & - G_{\text{floor}} - G_{\text{small}} - G_{\text{res}} - G_{\text{coup}} - G_{\text{fb}}. \end{aligned}$$

For every $U \in [2^a, 2^b]$, the auxiliary prime satisfies $U < R \leq \kappa U$. Monotonicity of $\mathcal{T}$ and $\alpha > 0$ therefore transfer (34) to $X = \kappa U$, with the additional intercept cost $\alpha \log \kappa$.

The auxiliary prime need not have a fixed ratio to U; the interval guarantee alone suffices. The certificate rounds slopes down and intercept costs up, obtaining

$$\mathcal{T}(e^x) \geq A_i x - D_i - \gamma \log(q/Q_0), \qquad \gamma = 3/10,$$

on the indicated finite interval. It checks $\beta/4 < \gamma$. The term involving q records the possible loss relative to the lower band endpoint Q_0.

6.2 Eight finite refinements

Each affine bound applies on a finite interval of x. Beyond its right endpoint, keep its endpoint value: monotonicity of $\mathcal{T}$ makes that value a valid lower bound at all later scales. Taking the maximum of the available bounds and zero gives a nondecreasing envelope.

For each of eight refinement stages, subtract $\gamma \log(Q_1/Q_0)$ from the preceding envelope and clip at zero. This gives a band-uniform lower bound Φ, which can be inserted into the factorial gain (28). Repeating the determinant comparison produces the next stage. The initial stage uses $\Phi = 0$; every later stage is an implication from previously proved bounds, with no limiting or fixed-point assertion.

Write $L(x)$ for a certified piecewise affine selection from the final bounds at Q_0, including zero and frozen endpoint values. Then throughout the band,

$$\mathcal{T}(e^x) \geq L(x) - \gamma \log(q/Q_0). \tag{35}$$

The exact parameter and output files record all lines, stages, and selected segments. For the rest of the proof, the conclusion of the determinant argument is precisely (35).

7 Positive kernels and the quadratic norm series

The lower bound (35) is strongest at different scales in different bands. We average over the scale using a positive kernel, then use the norm Euler product to bound the same average from above. The resulting norm series is estimated by Poisson summation, following the approach discussed by michaelmross and Tao [10, 2].

7.1 A positive average of the determinant lower bound

Choose positive rational d, e and a nonzero integer polynomial P, and normalize

$$h(x) = Z^{-1} e^{-dx} P(e^{-ex})^2 \quad (x \geq 0), \qquad Z = \int_0^\infty e^{-dx} P(e^{-ex})^2 \, dx.$$

Then $h \geq 0$ and $\int h = 1$. Expand $h(x) = \sum_i c_i d_i e^{-d_i x}$, with $d_i = d + ie$. All c_i and Z are exact rationals. The coefficients c_i may have either sign, but the unexpanded kernel is nonnegative. Integrating (35) gives

$$\int_0^\infty h(x) \mathcal{T}(e^x) \, dx \geq \sum_i c_i \int_0^\infty d_i e^{-d_i x} L(x) \, dx - \gamma \log(q/Q_0). \tag{36}$$

One and the same L is used in every signed term: the inequality is obtained by integration against $h \geq 0$, before expanding the square. Termwise lower estimates with unrelated functions L would not justify this step.

7.2 The prime weight and its Euler-product upper bound

Define

$$W(y) = \int_y^\infty h(x) f(e^{y-x}) \, dx.$$

This is nonnegative, nonincreasing, and at most 1. Tonelli's theorem and (13) show that

$$W(y) = \sum_i c_i F(d_i) e^{-d_i y}, \qquad F(d) = \int_0^1 d \, u^{d-1} f(u) \, du = \frac{2d\zeta(1+d)}{(1+d)(2+d)}. \tag{37}$$

The right side follows by integrating each $2d\,u^d(1-mu)_+$ from 0 to $1/m$. Thus the left side of (36) is $\sum_{p\ \mathrm{noninert}}(\log p)/p\,W(\log p)$.

We next express an upper bound for this weighted prime sum through the norm series. For $s>1$, put

$$\mathcal{A}_q(s)=\sum_{b\ge 1}\sum_{a\in\mathbb{Z}} Q_q(a,b)^{-s},\qquad B_q(s)=1+\mathcal{A}_q(s)/\zeta(2s).$$

Class number one and the unit group give $\zeta_K(s)=\zeta(2s)+\mathcal{A}_q(s)$. Its Euler product implies

$$-\frac{B_q'(s)}{2B_q(s)}=\sum_{p\ \mathrm{split}}\sum_{\substack{m\ge 1\\ m\ \mathrm{odd}}}\frac{\log p}{p^{ms}}+\frac{\log q}{2}\sum_{m\ge 1}(-1)^{m-1}q^{-ms}.$$

Multiply by the signed coefficients $c_iF(d_i)$ and sum at $s_i=1+d_i$. Since $W\ge 0$, the discarded higher odd powers have nonnegative weights. Adding $\frac{1}{2}\log q\sum_{m\ge 1}q^{-m}W(m\log q)$ also covers the ramified first-power term. It is at most

$$\frac{\log Q_0}{2(Q_0-1)}. \tag{38}$$

7.3 Poisson summation with an explicit error

The zero Fourier mode of the norm series has a closed form, and the remaining modes decay exponentially in v. Both the series and its derivative need explicit error bounds, because the Euler-product comparison uses a logarithmic derivative. Set

$$C(s)=\sqrt{\pi}\frac{\Gamma(s-1/2)}{\Gamma(s)},\quad g(s)=\psi(s)-\psi(s-1/2),\quad \mathcal{F}_q(s)=C(s)v^{1-2s}\zeta(2s-1).$$

Here $\psi=\Gamma'/\Gamma$ is the digamma function, distinct from the auxiliary function used in Section 5. Poisson summation in the real coordinate gives $\mathcal{F}_q$ as the zero-mode contribution. We bound the error uniformly on the band, allowing for the magnitudes of all signed kernel coefficients.

Choose any rational $0<a_0<v_-$. For $f_{s,t}(x)=(x^2+t^2)^{-s}$, shift the Fourier contour by $y=t-a_0$, where $t=bv$. Since

$$|(x+iy)^2+t^2|\ge x^2+a_0^2,$$

the nonzero Fourier coefficients are bounded by $C(s)a_0^{1-2s}e^{-2\pi n(t-a_0)}$. Summing both signs and all $b,n\ge 1$, put

$$z=e^{-2\pi v_-},\quad y_0=e^{-2\pi(v_--a_0)},\qquad E_A=\frac{2C(s)a_0^{1-2s}y_0}{(1-z)(1-y_0)}.$$

Then $|\mathcal{A}_q(s)-\mathcal{F}_q(s)|\le E_A$ uniformly in the band.

For the s-derivative, the complex argument of $(x+iy)^2+t^2$ has absolute value at most $\pi/2$, and its logarithm is bounded in modulus by

$$\log(1+x^2/a_0^2)+2\log(2t)+2|\log a_0|+\pi/2.$$

The integral of the first term against $(x^2+a_0^2)^{-s}$ is $C(s)a_0^{1-2s}g(s)$. Using $\log b\le b-1$ in the remaining geometric sum therefore gives

$$|\mathcal{A}_q'(s)-\mathcal{F}_q'(s)|\le E_N,\quad E_N=E_A\left\{2\log(2v_+)+2|\log a_0|+g(s)+\frac{\pi}{2}+\frac{2z}{1-z}\right\}. \tag{39}$$

The contour shift is justified by decay at infinity for $s>1$; the same decay with an extra logarithm justifies differentiation.

7.4 The resulting upper bound

Put $d = s - 1$, $z_2 = \zeta(2s)$, $c_s = -2\zeta'(2s)/\zeta(2s)$, and

$$\kappa_d = g(s) - 2\frac{\zeta'(2s-1)}{\zeta(2s-1)} - c_s.$$

The main upper term is

$$H_q(d) = \frac{\mathcal{F}_q(s)\{\log(q/4) + \kappa_d\}}{2\{z_2 + \mathcal{F}_q(s)\}}. \tag{40}$$

The certificate checks that $H_q(d) > 0$ and decreases with q for each exponent used. Since $z_2 > 1$, the logarithmic-derivative error is at most

$$E_H(d) = \frac{E_N + (c_s + 2H_{Q_0}(d))E_A}{2(1 - E_A)}. \tag{41}$$

Consequently a uniform upper bound for the weighted noninert sum is

$$U(q) = \sum_i c_i F(d_i) H_q(d_i) + \sum_i |c_i| F(d_i) E_H(d_i) + \frac{\log Q_0}{2(Q_0 - 1)}. \tag{42}$$

The main terms retain their signs. The error terms use absolute coefficient values, so cancellation is not assumed in the error estimate. Equations (36) and (42) now bound the same positive average from opposite sides.

8 Exact certification and completion of the proof

The final step has two numerical obligations on each band: a strict gap at Q_0, and a derivative comparison ensuring that the gap persists up to Q_1. All parameters are fixed before checking either obligation.

8.1 A contradiction on each whole band

The certificate uses the five bands

$$[570, 580], \quad [580, 600], \quad [600, 700], \quad [700, 1050], \quad [1050, 2000].$$

The first uses a polynomial of degree 16 with $d = 3/10, e = 1/25$; the others use degree 12 with $d = 2/5, e = 1/20$. The integer polynomial coefficients, auxiliary-prime rows, and eight-stage refinement schedule are fixed in `parameters.json`.

The lower bound (36) decreases with slope $-\gamma$ as a function of $Y = \log(q/Q_0)$. To compare the upper bound with it, write $a_i = 1/2 + d_i$ and $r_i = \mathcal{F}_{Q_0}(s_i)/\zeta(2s_i)$. Expand

$$H_q(d_i) = \frac{1}{2}(\log(Q_0/4) + Y + \kappa_{d_i}) \sum_{\ell \geq 1} (-1)^{\ell+1} r_i^\ell e^{-\ell a_i Y}.$$

The certificate bounds the differentiated geometric tail explicitly. For each retained ℓ, the signed finite sum is evaluated as a polynomial in $1 - e^{-\ell e Y}$, with its coefficients computed by exact interval arithmetic. This avoids naive interval cancellation. Cells of width $1/1000$ cover the entire Y-interval. On every cell the derivative of the main sum in (42) is strictly less than $-\gamma$. The two error bounds in that equation are uniform constants, so they require no differentiation.

Band	Lower bound	Upper bound	Gap >	Norm error <
$[570, 580]$	10.975413	10.965024	0.010389	7.652×10^{-5}
$[580, 600]$	9.566392	9.544454	0.021938	1.969×10^{-10}
$[600, 700]$	9.755174	9.682808	0.072367	4.675×10^{-11}
$[700, 1050]$	9.890348	9.507108	0.383240	6.739×10^{-14}
$[1050, 2000]$	12.460477	11.013702	1.446776	3.672×10^{-22}

All displayed bounds are rounded outward. The upper bound includes both the norm-series error and the ramified-prime allowance. The exact certificate endpoints are recorded in the JSON output.

At $q = Q_0$, the exact lower endpoint of (36) is strictly greater than the exact upper endpoint of (42). Since the latter decreases faster than $-\gamma \log(q/Q_0)$, (35) proves a contradiction throughout the band. The five bands exclude $[570, 2000]$; the audited v5 implication excludes $q \geq 2000$, proving Theorem 1.

8.2 The remaining finite range

The analytic cutoff reduces the class-number-one problem to a short finite check. We include it separately, following the small-prime sieving suggestion in Pratt's comment [8]. For $q > 8$, the same norm obstruction used in Section 2 forces $q \equiv 3 \pmod 8$, and every prime $p < q/4$ must be inert. The program `verify_finite.py` tests these necessary conditions on the 70 integers $q = 11, 19, \ldots, 563$. Primes at most 109 suffice to leave only $11, 19, 43, 67, 163$. Together with the fundamental discriminants of absolute value at most 8, the survivors are $3, 4, 7, 8, 11, 19, 43, 67, 163$.

The program also enumerates reduced positive definite primitive binary quadratic forms of discriminant $-q$ for these nine values and finds exactly one in each case. This uses the standard correspondence between their classes and ideal classes. The finite check completes the classification below the cutoff; it plays no role in proving Theorem 1.

8.3 Reproducing the certificate

The numerical proof uses integer and rational intervals throughout. Logarithms use a convergent positive series after range reduction; exponentials use a Taylor series with an explicit tail; square roots use integer square roots. Zeta values and derivatives use Euler–Maclaurin with the periodic-Bernoulli remainder, while gamma and digamma use positive-real Stirling expansions with remainder bounds. These are evaluations in $\Re s > 1$, not zero-location inputs. Appendix A specifies the remainder formulas. Floating-point searches selected the fixed inputs, but acceptance depends only on the stated interval inequalities.

Run, without Python's `-O` option,

```
python3 verify_v6.py
python3 verify_structures.py
python3 verify_finite.py
```

The first command certifies the inequalities and writes exact endpoints to JSON. The second checks finite instances of the new algebraic identities and divisibility statements; these tests do not replace their proofs. The third performs the finite check in Section 8.2.

The recorded run passes 429 grouped checks, including 1,258 cells for the dependence on q, using 184 auxiliary-prime parameter rows. Its smallest strict gap exceeds 0.010389. The parameter file and exact output files are part of the accompanying package. The programs check explicit inequalities; they do not construct the theoretical determinant.

Acknowledgements and AI disclosure

We thank Tao for organizing the challenge and the participants for sharing their arguments publicly. Their contributions are cited where they enter the proof. Green's paper attributes its mathematical arguments to ChatGPT and its exposition to Green. No priority claim is made for the refinements in this note.

Codex developed the refinements, selected parameters, and prepared the manuscript and programs. Independent mathematical review of this v6 draft is pending. Exact numerical certification and finite algebraic tests supplement mathematical review; this is not formal verification. The accompanying `RESEARCH.md` records alternative approaches that were explored. The cutoff 570 is a stopping point for this research pass, not an optimality theorem about the method.

A Numerical remainder bounds

Here are the remainder formulas used in the certificate; the integer parameters in this paragraph are numerical evaluation parameters, independent of the determinant dimensions. For $s > 1$, Euler–Maclaurin with $N_0 = 64$, $m = 18$ gives

$$\begin{aligned}
\zeta(s) &= \sum_{n<N_0} n^{-s} + \frac{N_0^{1-s}}{s-1} + \frac{1}{2}N_0^{-s} + \sum_{k=1}^{m} \frac{B_{2k}(s)_{2k-1}}{(2k)!} N_0^{1-s-2k} + \mathcal{R}, \\
|\mathcal{R}| &\le E = \frac{|B_{2m}|(s)_{2m}}{(2m)!(s+2m-1)} N_0^{1-s-2m}, \\
|\mathcal{R}'| &\le E\left\{\log N_0 + \sum_{j=0}^{2m-1} \frac{1}{s+j} + \frac{1}{s+2m-1}\right\}.
\end{aligned}$$

Here $(s)_j$ denotes the rising factorial. These bounds follow from the periodic-Bernoulli integral and its derivative. For log gamma and digamma, shift to $z = x + 64$, retain the terms $1 \le k < 20$ in the positive-real Stirling expansions, and use the first omitted term as an absolute remainder bound [13]:

$$|R_{\log\Gamma}| \le \frac{|B_{40}|}{40 \cdot 39\, z^{39}}, \qquad |R_{\psi}| \le \frac{|B_{40}|}{40 z^{40}}.$$

The recurrence formulas then recover the values at x.

After range reduction to $1 \le y \le 2$, logarithms use 80 terms of $2\sum_{k\ge 0} t^{2k+1}/(2k+1)$, where $t = (y-1)/(y+1)$, and tail $2t^{161}/(161(1-t^2))$. Exponentials are reduced to $0 \le x \le 1$; the series through degree 80 has positive tail at most $82/(81 \cdot 81!)$. Repeated squaring and reciprocation undo the reduction. Machin's formula with 60 arctangent terms encloses π. Elementary interval operations round outward to 65 decimal places; square roots use integer bounds.

For the differentiated geometric series in Section 8, the tail after L terms is at most

$$\frac{1}{2}\sum_i |c_i| F(d_i) r_i^{L+1} \left\{\frac{1}{1-r_i} + a_i E_* \left(\frac{L+1}{1-r_i} + \frac{r_i}{(1-r_i)^2}\right)\right\}, \qquad E_* = \max_i \{\log(Q_1/4) + \kappa_{d_i}\}.$$

The certificate increases L until this bound is below 10^{-10}, then includes it in every derivative cell test.

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[13] NIST Digital Library of Mathematical Functions, Section 2.10 (Euler–Maclaurin) and Section 5.11 (Stirling expansions and positive-real remainder bounds).